Session 2: Spatial foundations, data acquisition and quality

Modeling the distribution of biodiversity under global climate change

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Principles of Geographic Information Systems (GIS)

What is GIS? Why does it matter for modeling biodiversity?

Definition: a system to capture, store, manipulate, analyze, and present spatial or data.

Relevance in macroecology: mapping habitats, tracking species distributions under climate change, analyzing environmental patterns (e.g., temperature), conservation planning (MPAs).

Specific link to SDMs: supports the representation of species locations and environmental variables; crucial for data preparation, interpretation and visualization.

Representing Reality: Vector Data Models

Represents **discrete features with distinct boundaries**.

Primitives:

- **Points:** x, y (,z) locations (e.g., occurrences, stations).
- **Lines:** points connected by segments (e.g., coastlines, tracks, currents).
- **Polygons:** closed sequence of segments representing areas (e.g., MPAs, islands).

Features are linked to an attribute table (e.g., species name, MPA size).

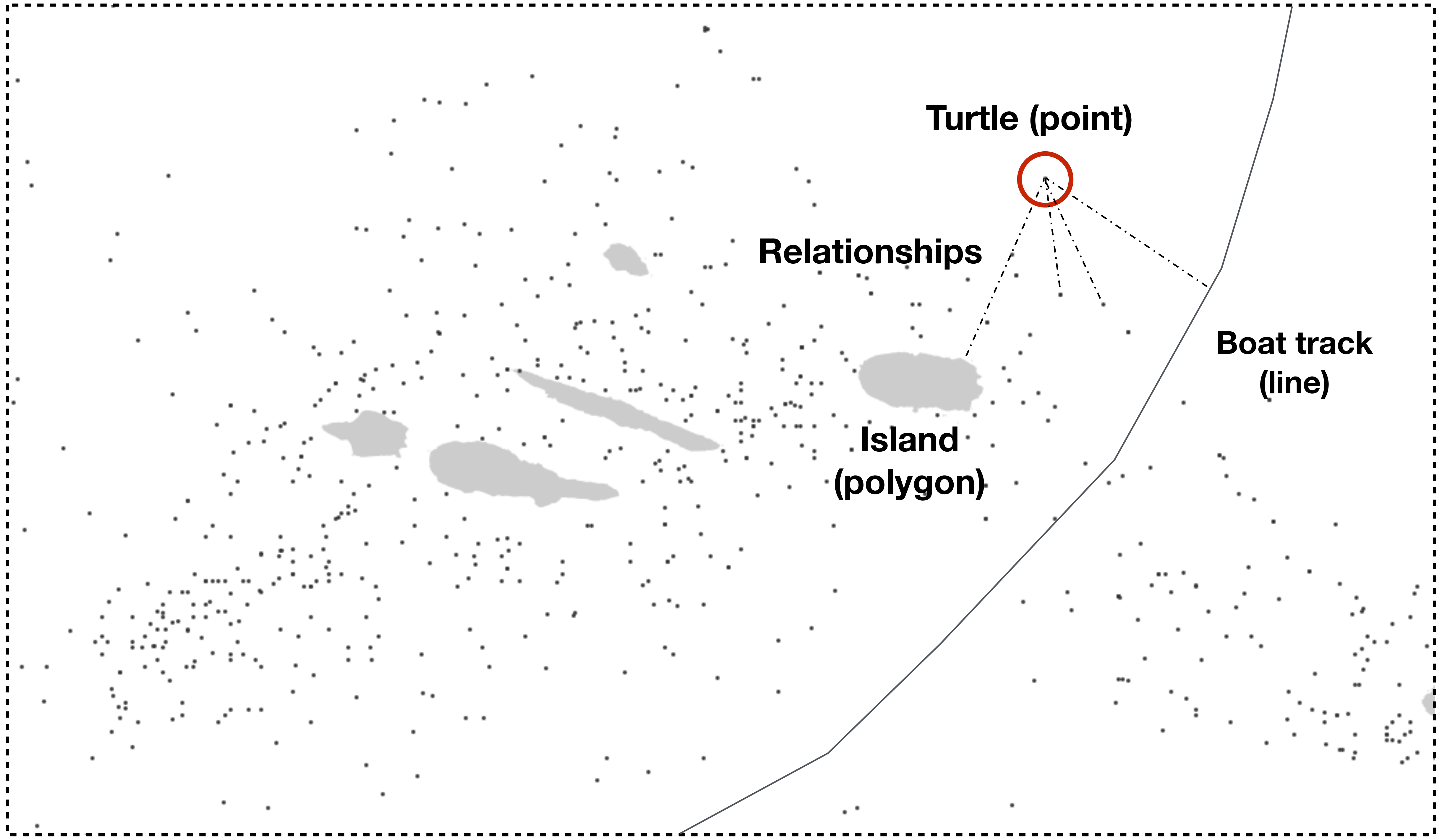
Can store spatial relationships (e.g., distances between sites).

Spatial information for *Caretta caretta*

Locations		Attributes						Relationships
Lon	Lat	Species	Common name	Date	Depth	Behaviour	Observer	Distance to land
0,61287	39,58955	Caretta caretta	Loggerhead sea turtle	10/08/12	22	Swimming	John Doe	232
0,83999	39,77417	Caretta caretta	Loggerhead sea turtle	03/03/12	12	Swimming	John Doe	3454
3,18192	42,40479	Caretta caretta	Loggerhead sea turtle	14/08/11	1	Eating	Unknown	23
-4,86395	55,24295	Caretta caretta	Loggerhead sea turtle	01/01/04	0	Swimming	Unknown	2567
-5,89297	57,06977	Caretta caretta	Loggerhead sea turtle	03/03/12	0	Sleeping	John Doe	878
34,9082	32,55822	Caretta caretta	Loggerhead sea turtle	22/12/02	23	Sleeping	John Doe	777
0,61287	39,58955	Caretta caretta	Loggerhead sea turtle	14/08/11	12	Swimming	John Doe	6
0,83999	39,77417	Caretta caretta	Loggerhead sea turtle	03/03/12	2	Sleeping	John Doe	78

Vector data model

Points, lines and polygons (locations, and features with attributes)



The distribution of the Loggerhead sea turtle in the Azores Islands

Common file formats

Vector data are common as **shapefiles** (a standard since 1990).

.shp is the geometry of vector features

.dbf are the attributes of vector features

.shx helps the GIS Application to find features more quickly.

Vector data are migrating to simpler formats like **GeoPackage files**, which are an open, portable, self-describing, compact formats for spatial information.

.gpkg is the geometry and the attributes of vector features.

saccorhiza

ID	SPECIES	Y	LON	LAT	REF
1	Sacchorhiza polyschides	1966	-5.600819	36.018775	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
2	Sacchorhiza polyschides	1993	-5.601822	36.019239	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
3	Sacchorhiza polyschides	1960	-5.653864	36.062617	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
4	Sacchorhiza polyschides	1965	-5.44705	36.116542	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
5	Sacchorhiza polyschides	1958	-5.353586	36.14075	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
6	Sacchorhiza polyschides	1965	-5.922481	36.190028	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
7	Sacchorhiza polyschides	1966	-6.087778	36.277683	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
8	Sacchorhiza polyschides	1983	-5.158386	36.422061	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
9	Sacchorhiza polyschides	1982	-4.717386	36.493706	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
10	Sacchorhiza polyschides	1982	-4.882289	36.510117	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
11	Sacchorhiza polyschides	1982	-4.622617	36.571558	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
12	Sacchorhiza polyschides	1966	-4.421261	36.721283	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
13	Sacchorhiza polyschides	1970	-8.936489	37.009864	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
14	Sacchorhiza polyschides	1963	-8.978044	37.023033	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
15	Sacchorhiza polyschides	1963	-8.667864	37.079811	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
16	Sacchorhiza polyschides	1963	-8.266667	37.083333	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
17	Sacchorhiza polyschides	1956	-8.45	37.083333	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX. Saccor
18	Sacchorhiza polyschides	1970	-8.247875	37.089072	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
19	Sacchorhiza polyschides	1970	-8.469325	37.098511	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
20	Sacchorhiza polyschides	1970	-8.896	37.184133	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX. Saccor
21	Sacchorhiza polyschides	1954	-8.770147	37.757611	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
22	Sacchorhiza polyschides	1970	-8.796936	37.906147	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
23	Sacchorhiza polyschides	1970	-9.216664	38.416647	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
24	Sacchorhiza polyschides	1979	-9.1176	38.43685	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX. Saccor
25	Sacchorhiza polyschides	1970	-9.101489	38.444469	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
26	Sacchorhiza polyschides	1989	-8.984847	38.469364	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
27	Sacchorhiza polyschides	1982	-8.990794	38.495931	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
28	Sacchorhiza polyschides	1970	-9.355975	38.689786	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX
29	Sacchorhiza polyschides	1989	-9.483333	38.7	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX. Saccor
30	Sacchorhiza polyschides	1951	-9.269286	38.702911	"JosÈ Luis Izquierdo; Tom·s Gallardo; Isabel PÈrez-Ruzafa (1995) Mapas de distribuciÛn de algas marinas de la PenÌnsula IbÈrica e Islas Baleares. IX

Common file formats

Vector data are also available as **text delimited files** (e.g., .txt, .csv), delimited by a **Tab, a comma or semicolon**. Useful for point data and matrices (e.g., excel).

Representing Reality: Raster Data Models

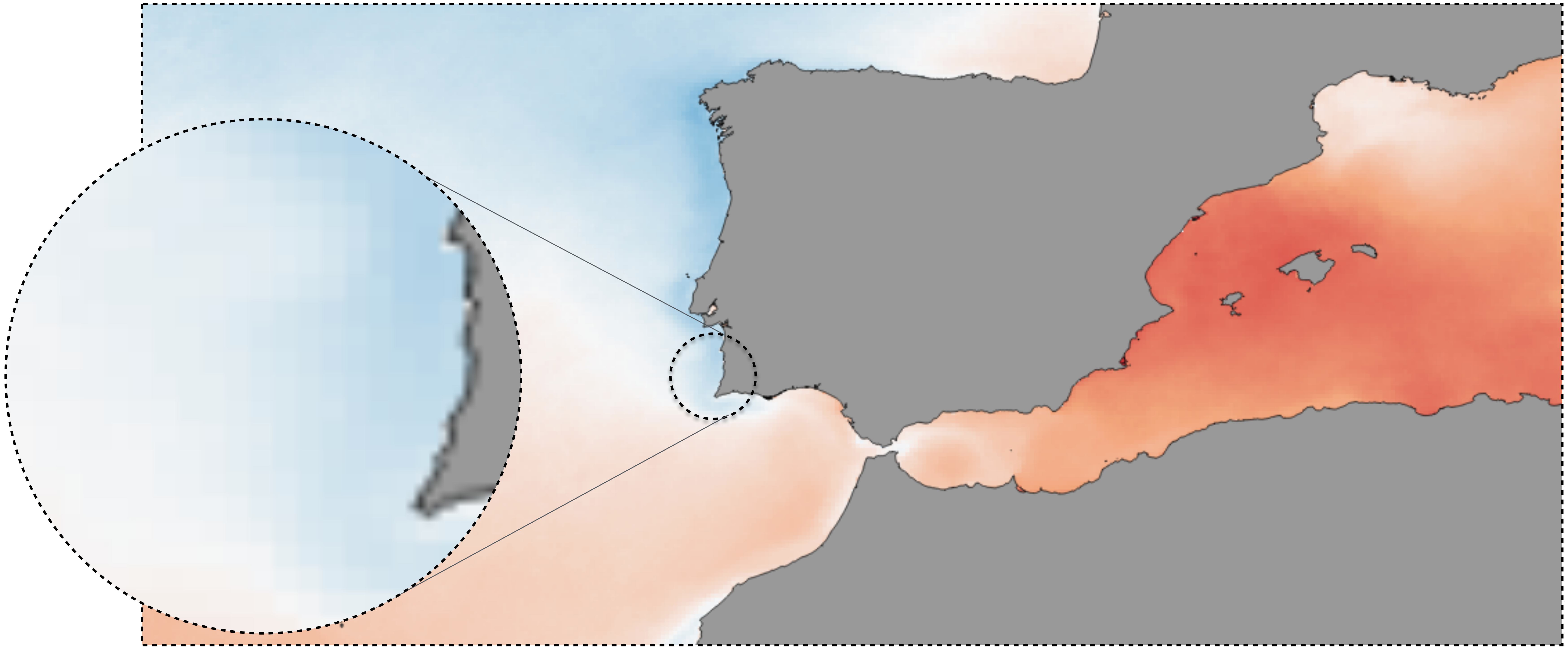
Composed of a **grid of cells** (pixels), where each cell **contains a value representing a specific feature** (e.g., temperature, depth, suitability score).

Raster data **can be continuous** (e.g., temperature) or **thematic** / categorical (e.g., habitat type).

SDMs rely heavily on digital raster layers to link biological occurrence data to extant environmental conditions

Raster data model

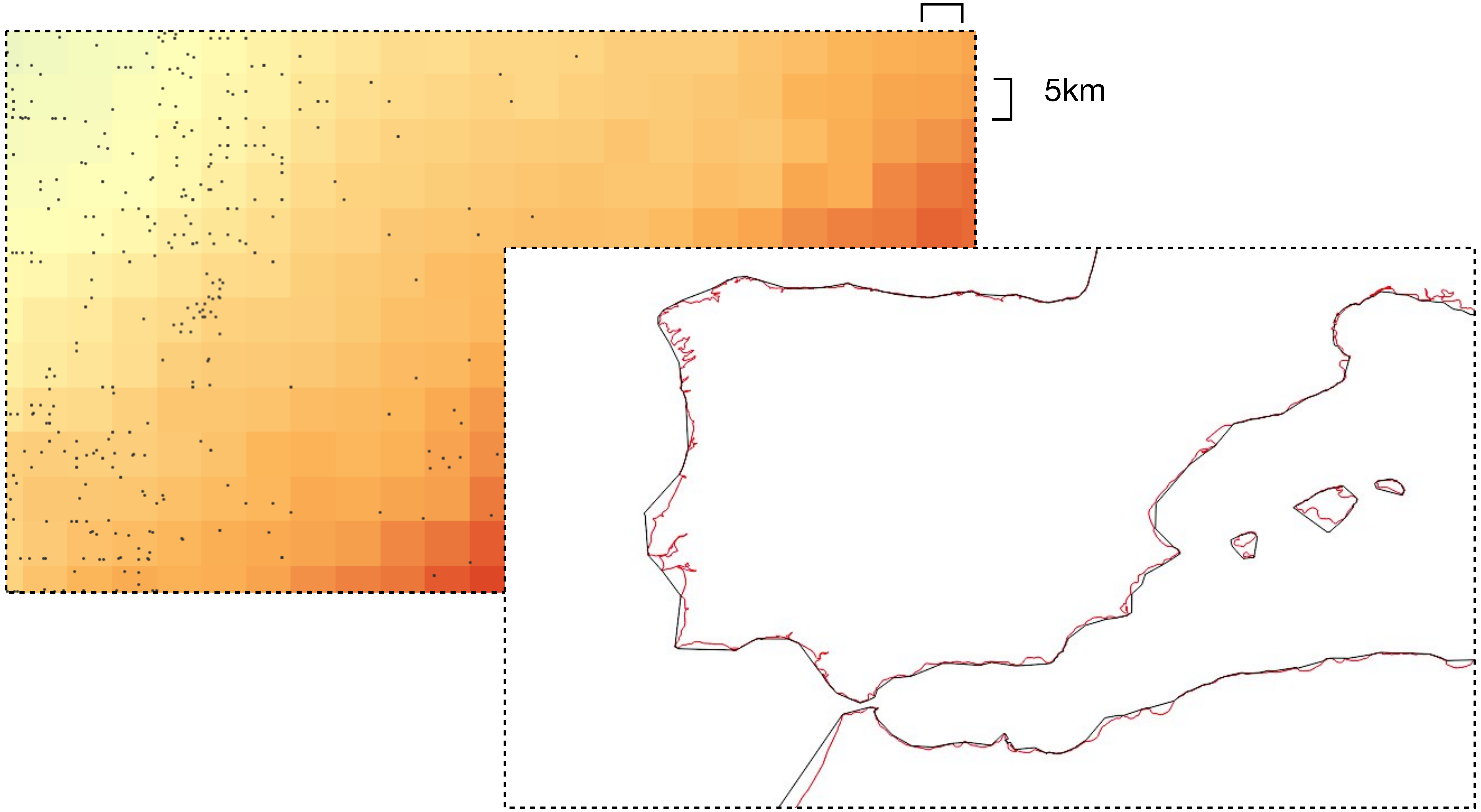
Grids and cells representing continuous features



SST in the western Mediterranean Sea and adjacent Atlantic Ocean.

Common file formats

Raster data are the the industry standard for GIS and satellite remote sensing applications. **Files are typically as .tif, .asc or .nc**.



Both models have well-defined spatial **extent and resolution**, which is the size of the **smallest feature able to be recognized.**

Vector data model: number of vertex;

Raster data model: size of cells;

Where on earth? Coordinate Reference Systems (CRS)

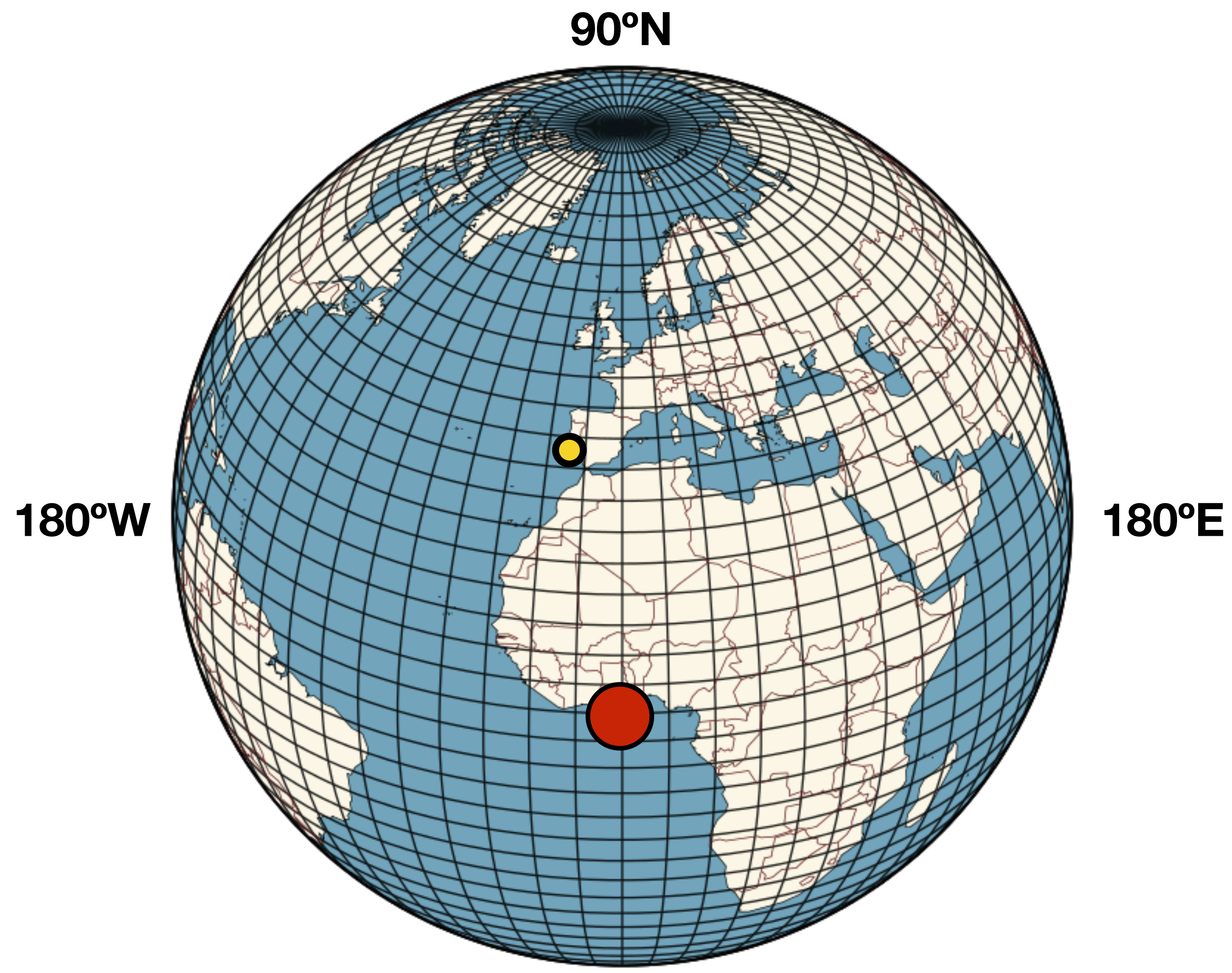
Coordinate reference systems **allow representing the locations of features within a common geographic framework**.

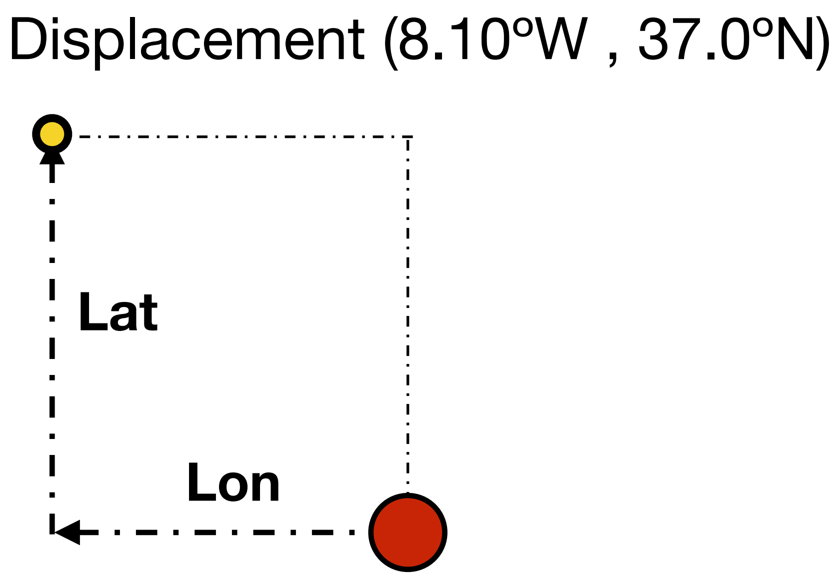
Without a CRS, coordinates are numbers that do not represent a specific place.

e.g., 7.321 ; -121.981

A CRS gives meaning to the information of location.

CRS have an inherent **fundamental challenge**: representing the 3D Earth on 2D surfaces.





Reference point (Datum) at the interception between:

Equator & Prime Meridian (0ºN 0ºE)

CRS type 1: Geographic Coordinate Systems (GCS)

Defines locations directly on the 3D Earth using latitude and longitude (angles).

Components:

- **Datum:** positions locations relative to the Earth.
- **Displacement units:** degrees (usually decimal degrees).

Decimal degrees are simpler to use.

37°2'38.08"N ; 7°58’20.73”W

37.043911° ; -7.972425° 👈🏼

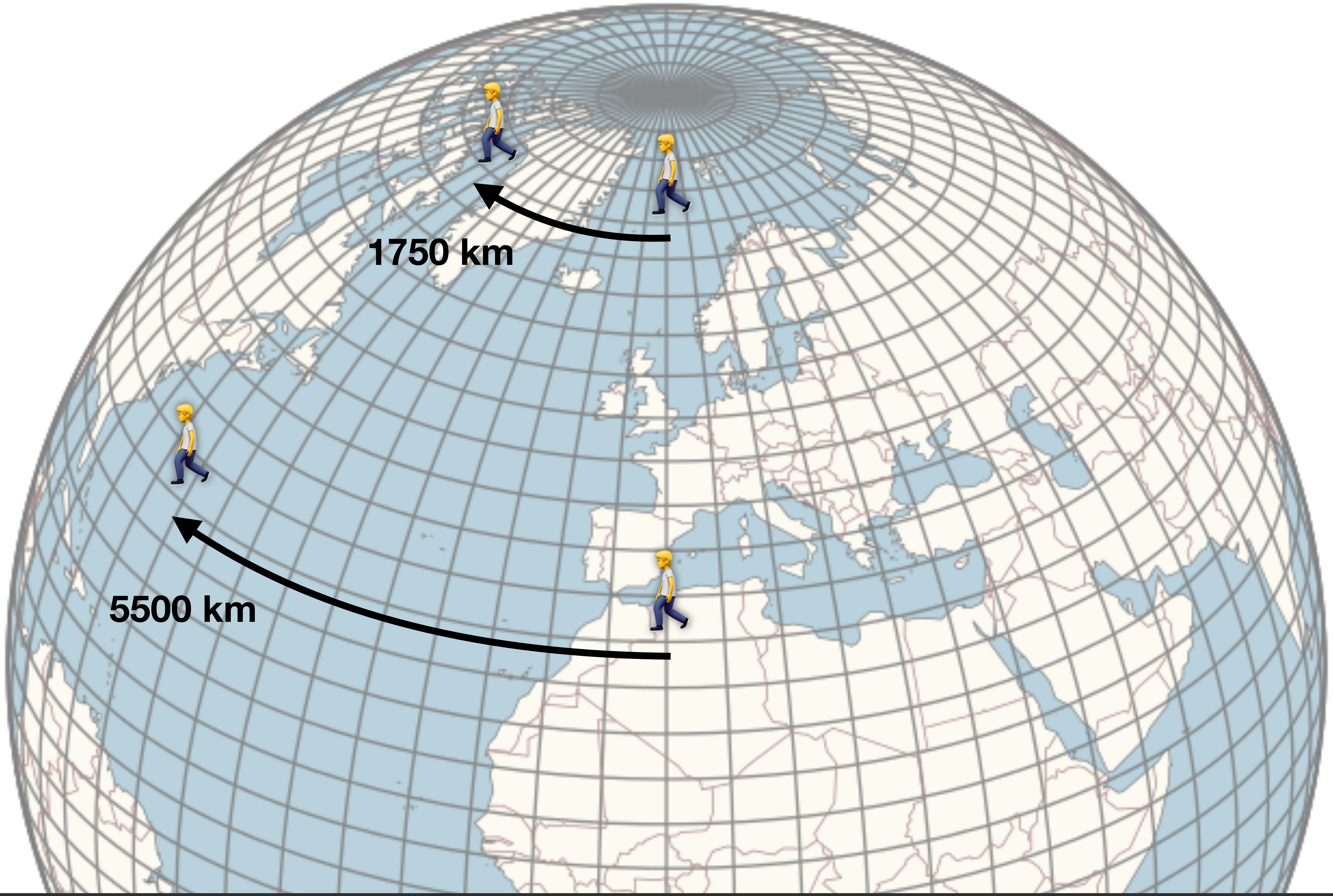
Simpler to type and share with others;

In data tables (e.g., R, Excel), longitude and latitude entries **use single cells, allowing direct analyses** (e.g., calculate distances).

Geographic coordinate systems

GCS **are not good to make measurements**.

There is the need to project data to an alternative coordinate system (e.g., based on actual distances).

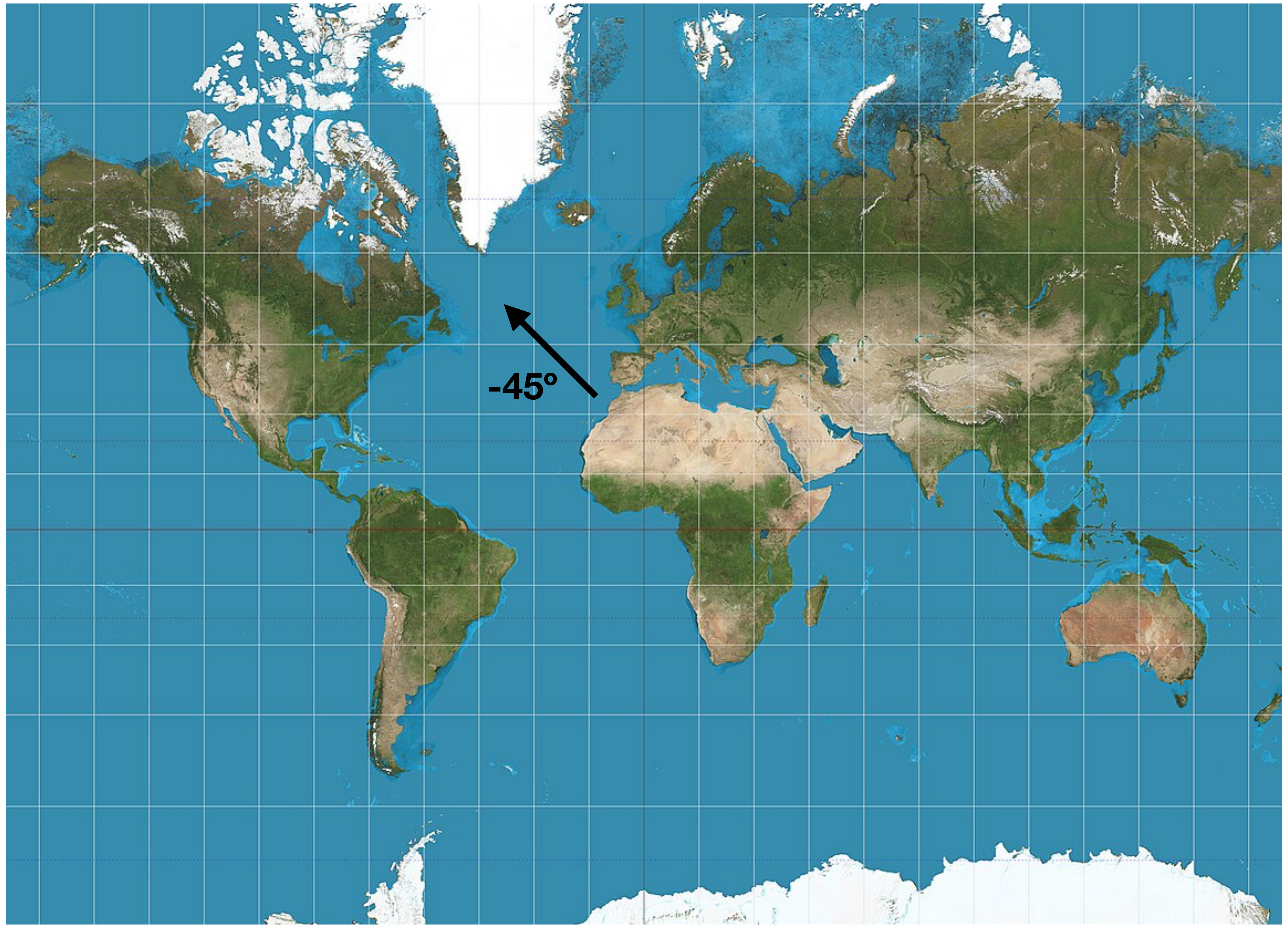


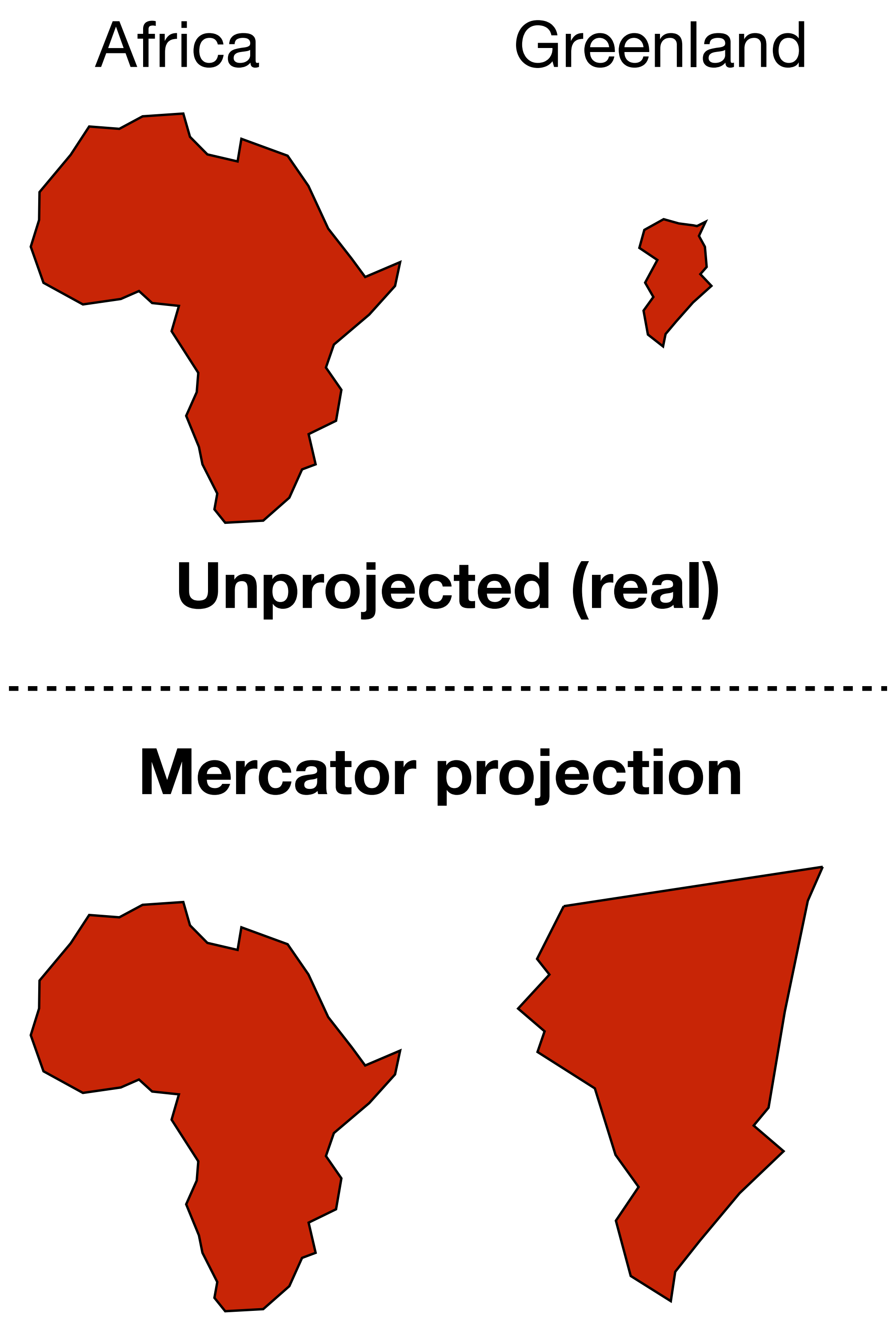
Displacement of 60º performed at different latitudes (360º longitude).

CRS type 2: Projected Coordinate Systems (PCS)

Mathematical transformation (**projection**) of the Earth's surface onto a 2-dimensional plane (**map**) with **constant units of distance (e.g., m), area, direction across the two dimensions**.

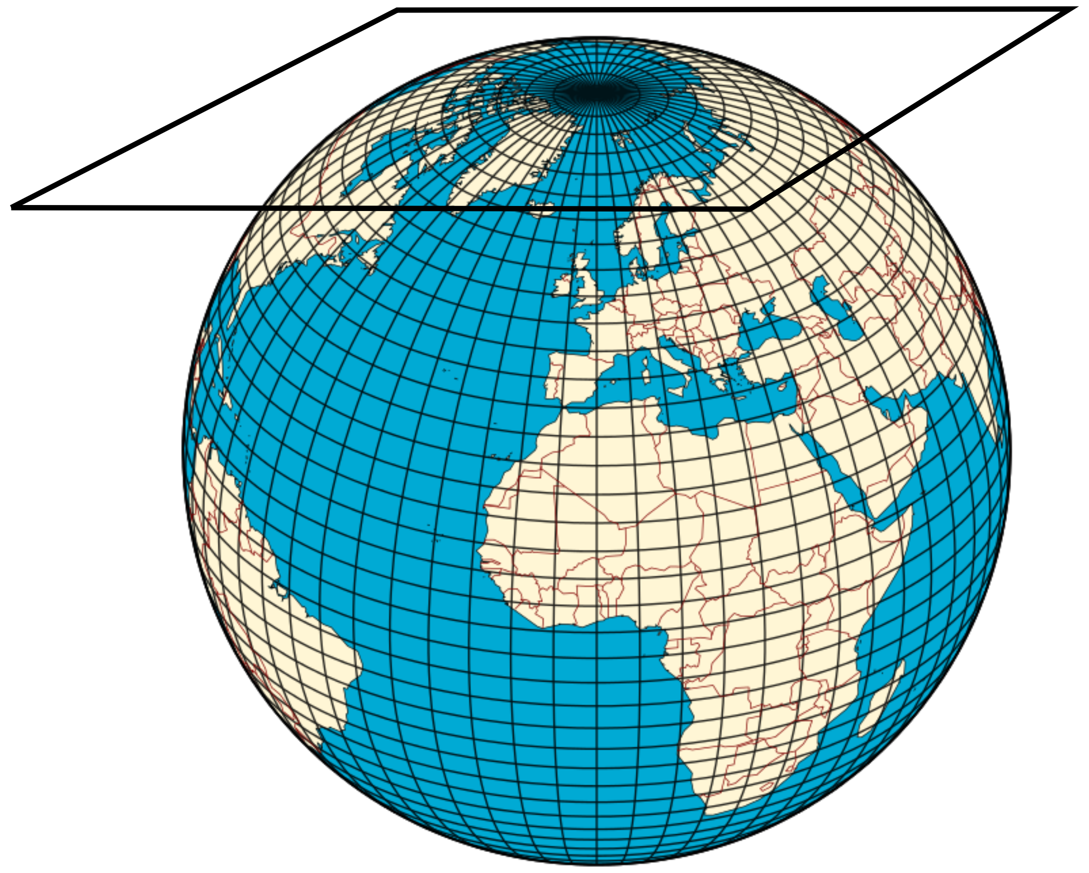
Crucially, **all map projections introduce distortion**. It's impossible to perfectly flatten a curved surface. Projections inevitably distort one or more properties.



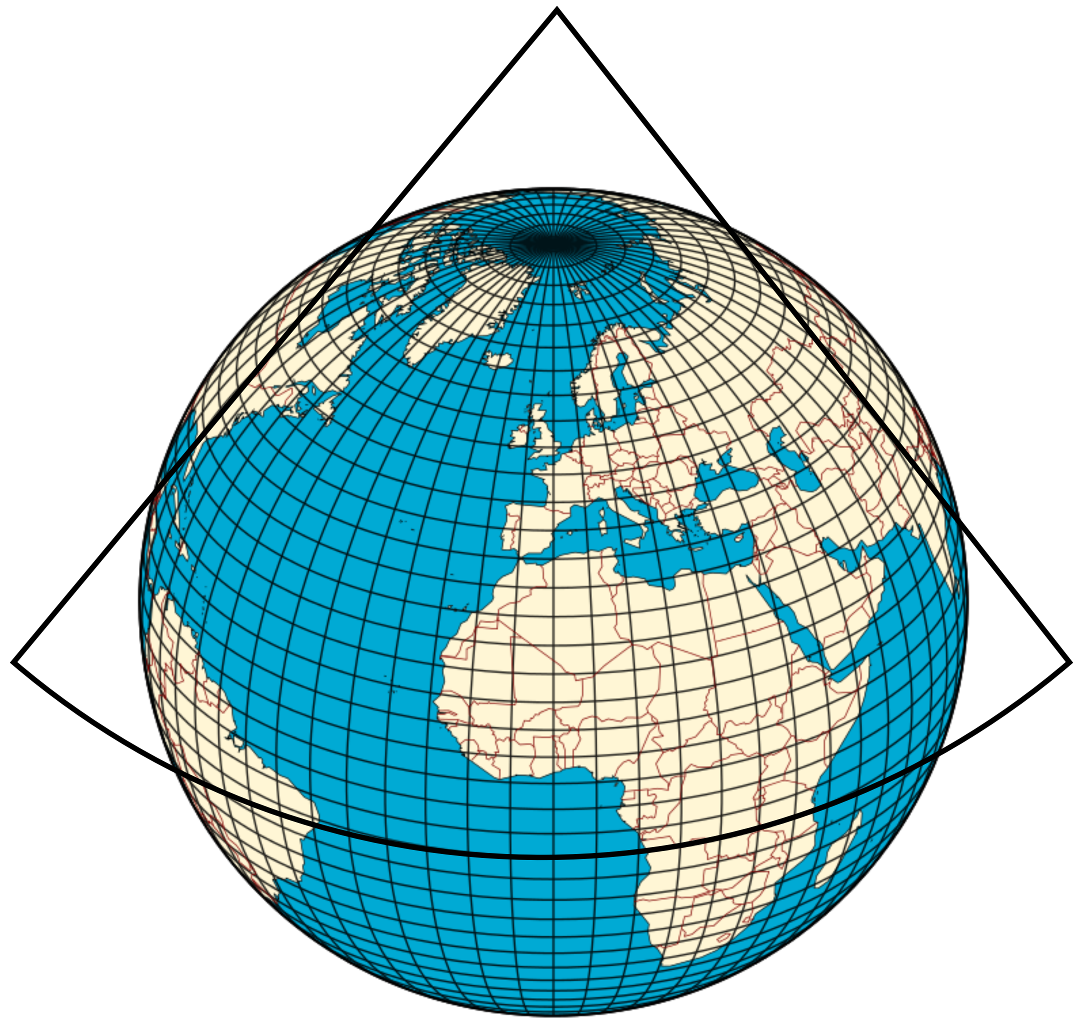


Mercator is a cylindrical projected coordinate systems that **represents lines of constant angle** (i.e., course; good for nautical use).

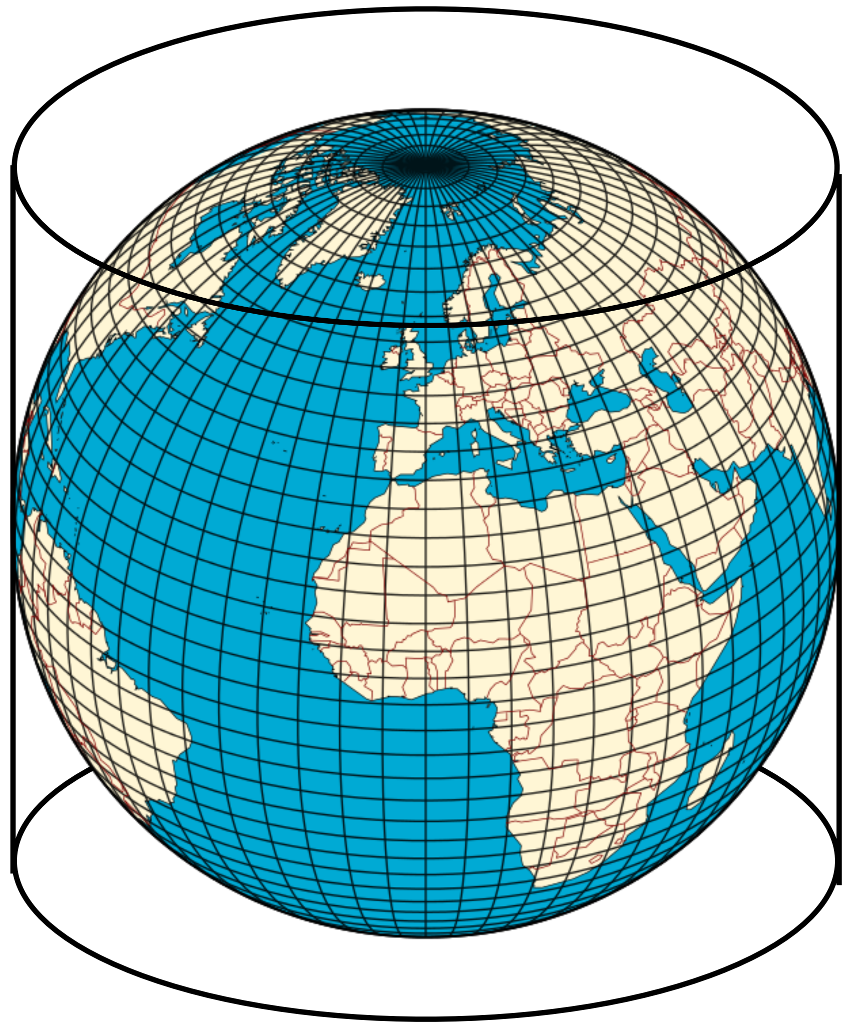
Mercator projection, distortion increases towards higher latitudes.



Planar projection
(poles)



Conical projection
(mid-latitudes)



Cylindrical projection
(rectangular maps)

There is no best projection. **Map-maker must decide which property is important** for the application (e.g., equal area for area calculations, equidistant for distance measurements).

What can GIS do in a SDM context?

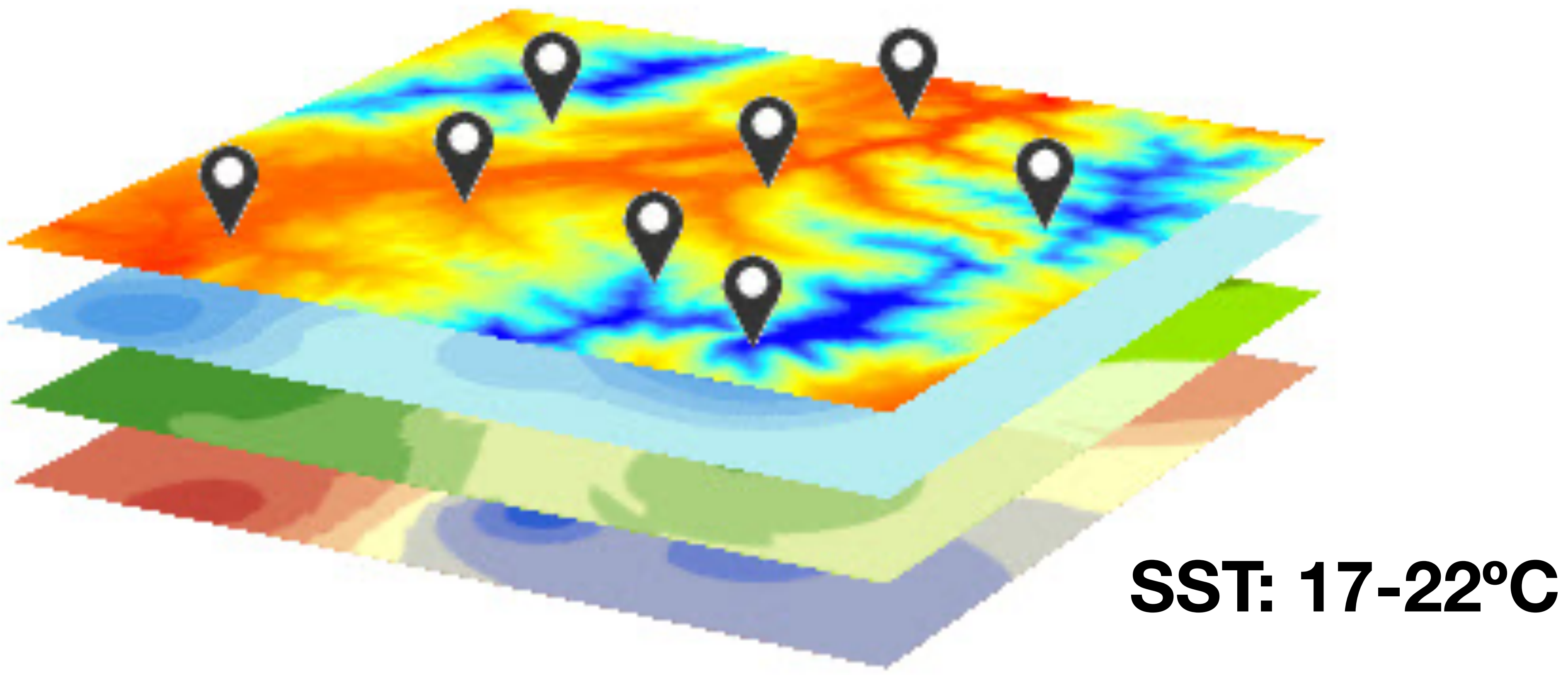
Data Preparation & Cleaning: e.g., species occurrence records (vectors) mistakenly recorded out of their known distribution.

Environmental Extraction: Using the geographic coordinates of species occurrences to **extract the exact environmental values** (e.g., temperature, precipitation) from underlying continuous raster layers to build the modeling datasets.

Spatial Cropping & Masking: Trimming global environmental raster layers **to the specific study area**, focusing the model on accessible habitats.

Mapping & Visualization: Converting complex, continuous mathematical model outputs into **visual maps** - such as habitat suitability heatmaps or binary presence/absence maps - that can inform conservation and policy.

Biological Data for SDMs



Occurrence records for modeling

Occurrence records allow building SDMs and answer *why* a species occurs in a given area.

Essential components: Must reference a particular taxon, at a particular place (geographic coordinates), and (ideally) at a particular time.

Metadata can describe the source of record (sampling method), means of identification (e.g., visual census), and spatial precision of record (maximum error).

Occurrence data for SDM: one species example

Data | **Metadata**

Lon	Lat	Date	Depth	Species	Common name	Behaviour	Observer
0,61287	39,58955	10/08/12	22	Caretta caretta	Loggerhead sea turtle	Swimming	John Doe
0,83999	39,77417	03/03/12	12	Caretta caretta	Loggerhead sea turtle	Swimming	John Doe
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34,9082	32,55822	22/12/02	23	Caretta caretta	Loggerhead sea turtle	Sleeping	John Doe
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0,83999	39,77417	03/03/12	2	Caretta caretta	Loggerhead sea turtle	Sleeping	John Doe
3,18192	42,40479	22/12/02	5	Caretta caretta	Loggerhead sea turtle	Eating	Unknown
-4,86395	55,24295	03/03/12	7	Caretta caretta	Loggerhead sea turtle	Swimming	Unknown
-5,89297	57,06977	01/01/04	3	Caretta caretta	Loggerhead sea turtle	Eating	Unknown

Where to Source Biological Data

Global Biodiversity information centres;

Peer-reviewed literature;

Reference collections (e.g. museums and herbaria);

Citizen science initiatives.

Occurrence records: Global Biodiversity information centers

(i.e., databases linked via a common access portal).

Global Biodiversity Information Facility (GBIF; https://www.gbif.org/):

Occurrence records: over 2.7 billion

Ocean Biodiversity Information System (OBIS; https://obis.org/):

Occurrence records: over 130 million

National Biodiversity Network (NBN; UK; https://nbnatlas.org/):

Occurrence records: over 320 million

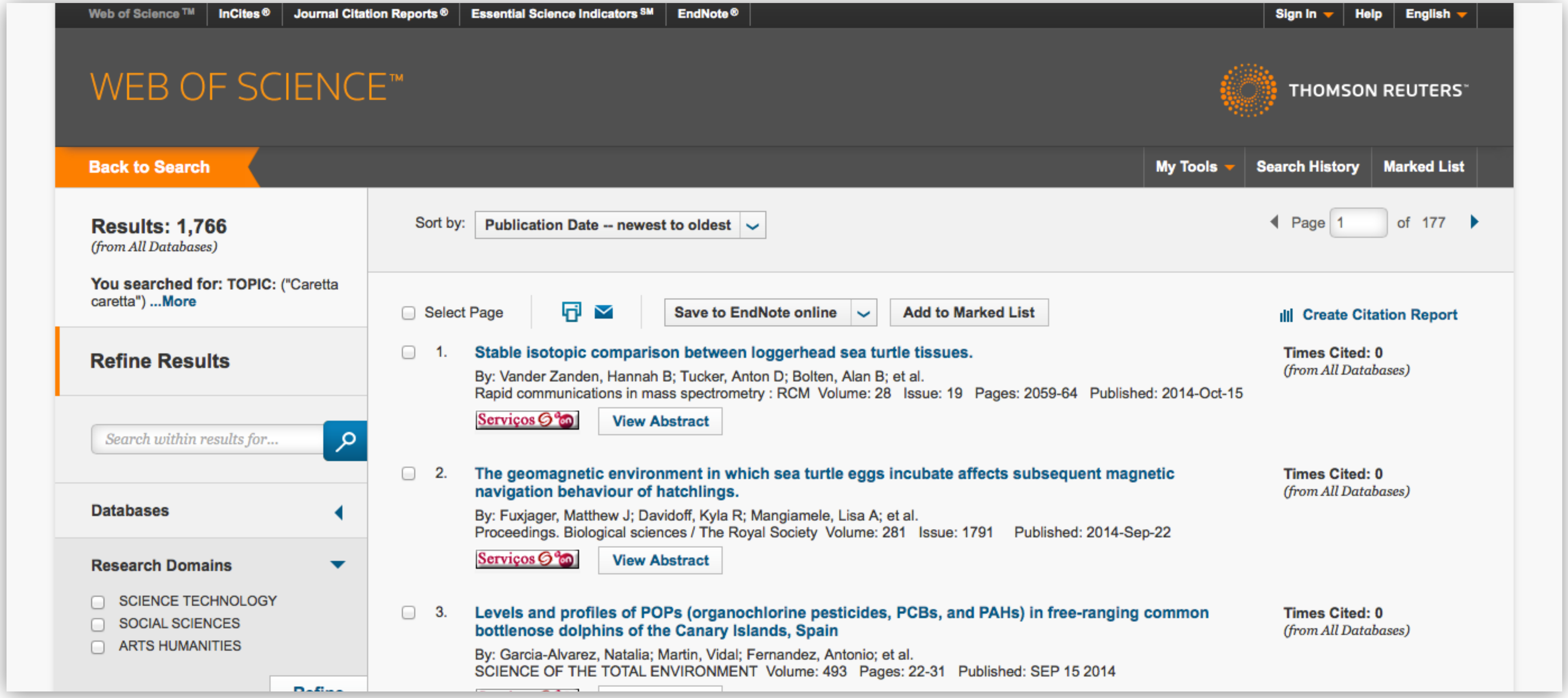
Atlas of Living Australia (ALA; https://www.ala.org.au/):

Occurrence records: over 130 million

Occurrence records: Literature

Literature is useful to compile biodiversity records.

Peer-reviewed publications are preferred, even when out of the scope of our work.



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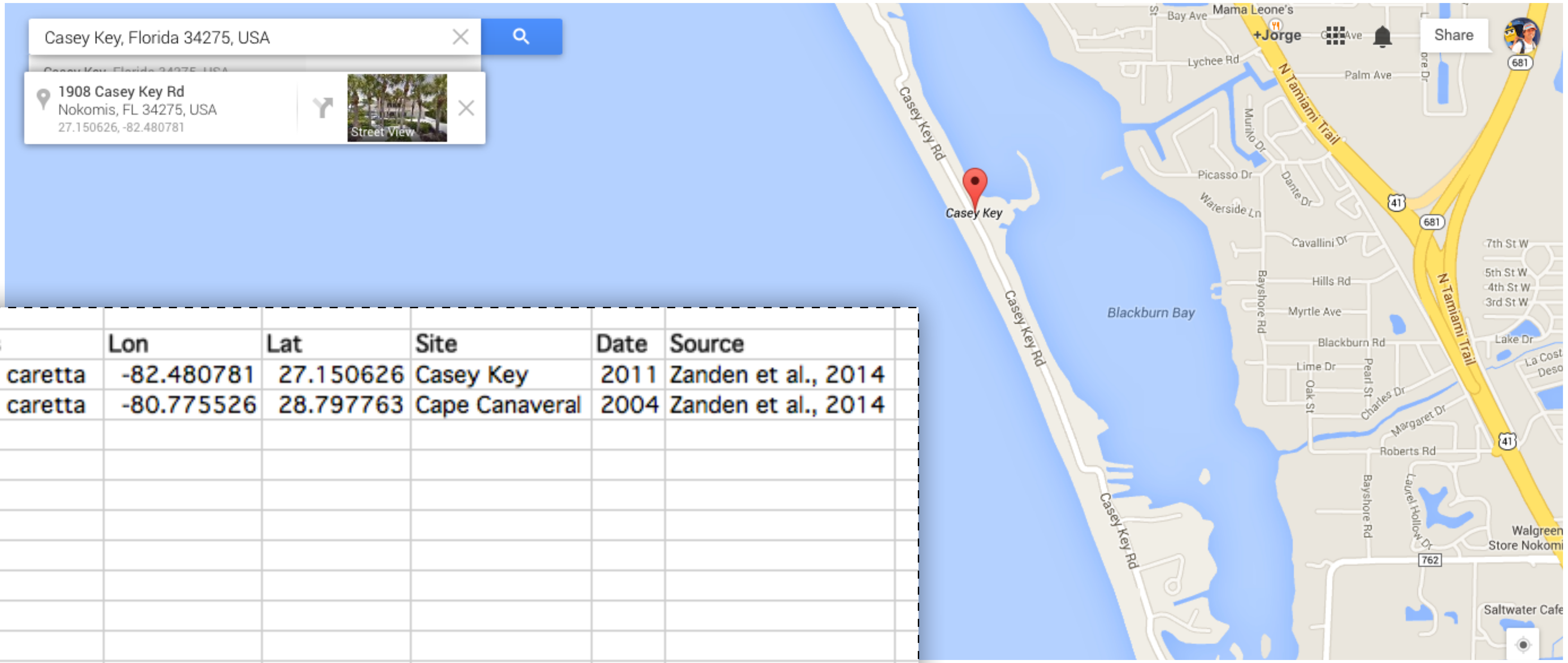


Species	Lon	Lat	Site	Date	Source
Caretta caretta	-82.480781	27.150626	Casey Key	2011	Zanden et al., 2014
Caretta caretta	-80.775526	28.797763	Cape Canaveral	2004	Zanden et al., 2014

Occurrence records: Biological reference collections

Preserved specimens / **photographs** maintained in **museums**, **universities**, **botanic gardens**, and similar **institutions**.

A major tools for basic investigation and assessment of biodiversity.

~2.5 billion specimens in preserved collections.

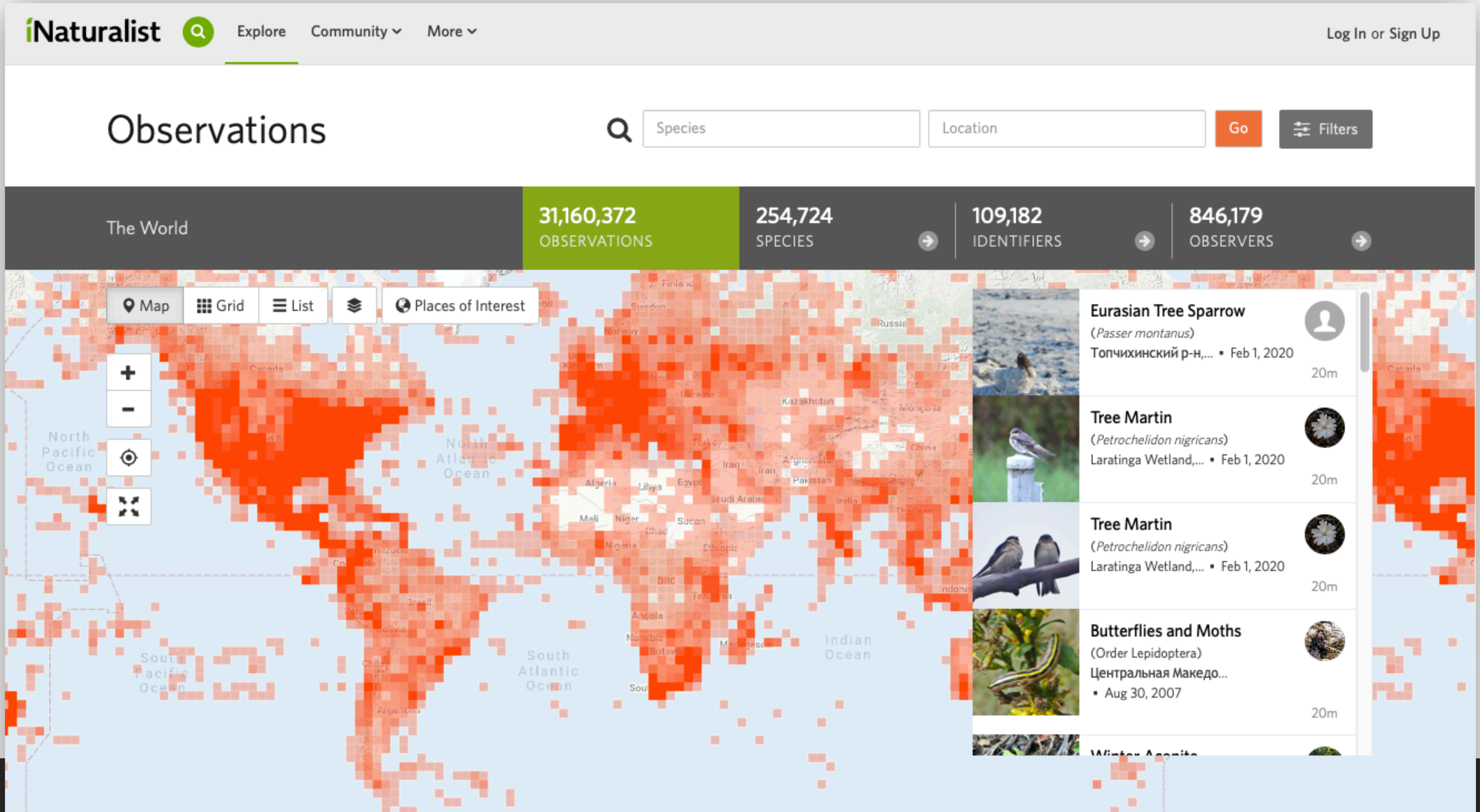
Macroalgae Herbarium Portal (www.macroalgae.org):

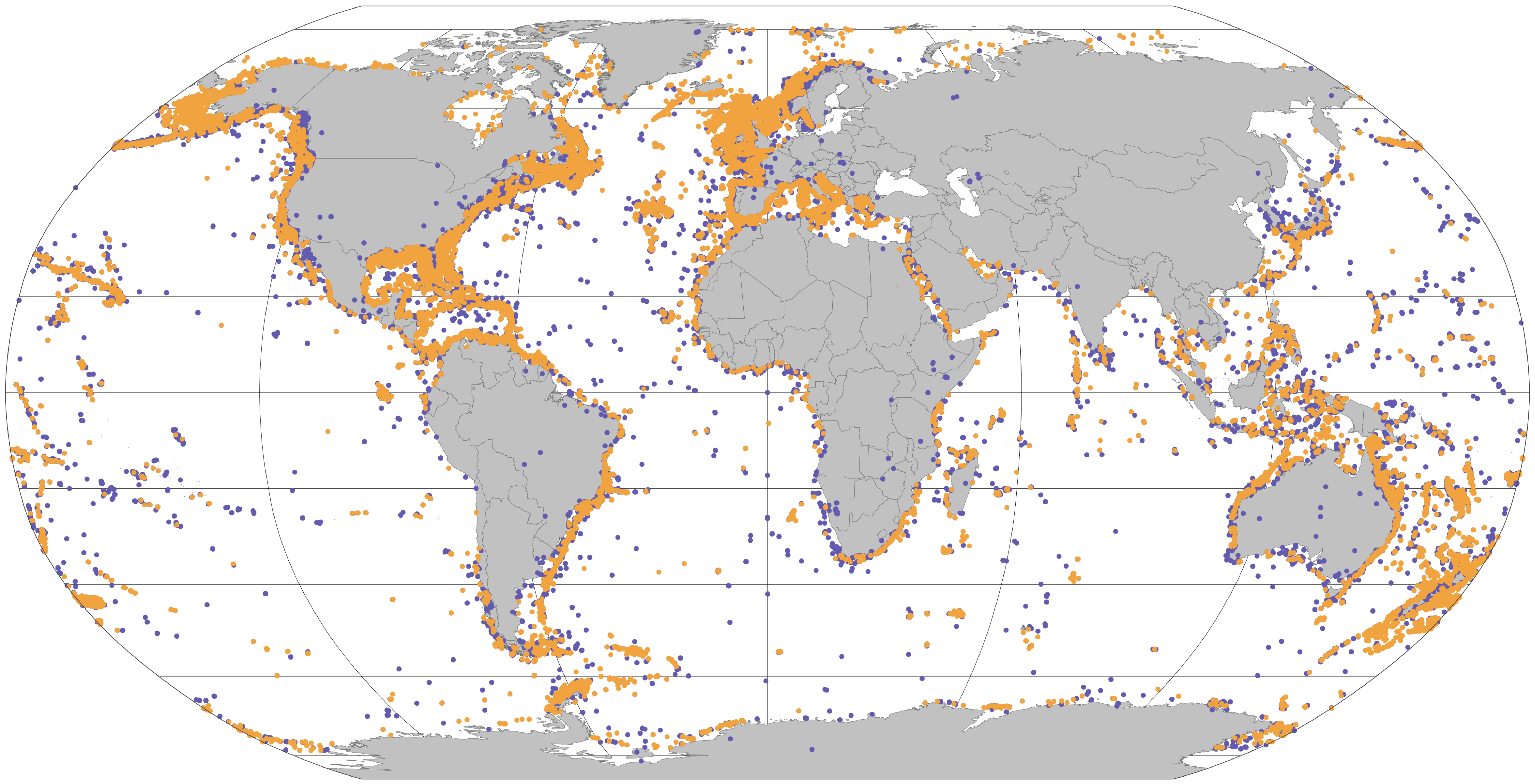
Occurrence records: >1 million records.

Occurrence records: Citizens can help

Sharing biodiversity records / pictures with scientists.

iNaturalist (http://www.inaturalist.com): > 300 million records.





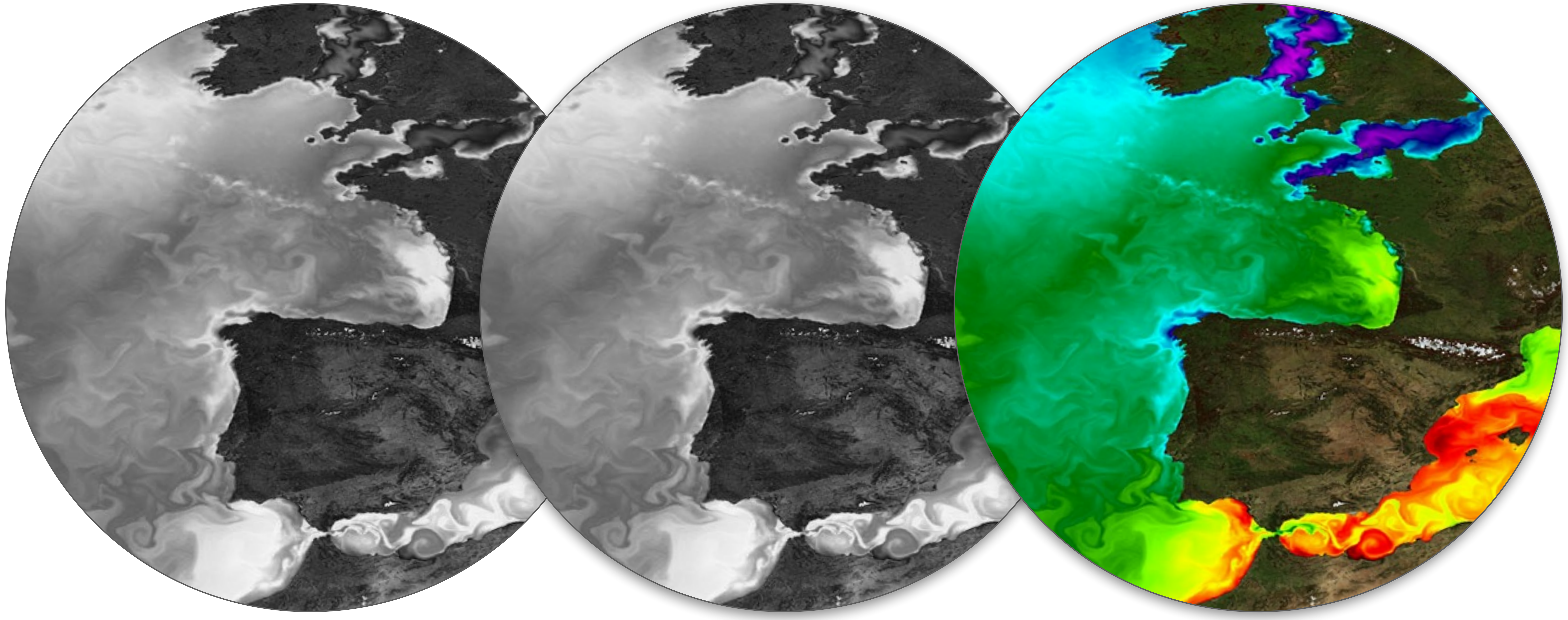
4 million records of ~1,500 species of kelp, seagrass and cold-water corals.

Data gathered from **literature** and major **biodiversity information facilities**.

Quality control of records based on published and expert knowledge to **eliminate entries outside the known vertical and geographic distribution of species.**

Assis et al., 2020

Environmental Layers for SDMs



Environmental data

Allows **linking biodiversity records with the conditions their were exposed to.**

Environmental predictors typically in the form of digital raster layers.

For the **terrestrial realm**, GIS databases like **World-Clim** provide uniform resolution and global coverage. Incorporates CMIP6 Earth System Models and Shared Socioeconomic Pathways (SSPs), making it the standard for terrestrial SDMs.

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(wileyonlinelibrary.com) DOI: 10.1002/joc.5086

RMetS
Royal Meteorological Society

WorldClim 2: new 1-km spatial resolution climate surfaces for global land areas

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[b] *Department of Environmental Science and Policy, University of California, Davis, CA, USA*

ABSTRACT: We created a new dataset of spatially interpolated monthly climate data for global land areas at a very high spatial resolution (approximately 1 km^2). We included monthly temperature (minimum, maximum and average), precipitation, solar radiation, vapour pressure and wind speed, aggregated across a target temporal range of 1970–2000, using data from between 9000 and 60 000 weather stations. Weather station data were interpolated using thin-plate splines with covariates including elevation, distance to the coast and three satellite-derived covariates: maximum and minimum land surface temperature as well as cloud cover, obtained with the MODIS satellite platform. Interpolation was done for 23 regions of varying size depending on station density. Satellite data improved prediction accuracy for temperature variables 5–15% (0.07–0.17 °C), particularly for areas with a low station density, although prediction error remained high in such regions for all climate variables. Contributions of satellite covariates were mostly negligible for the other variables, although their importance varied by region. In contrast to the common approach to use a single model formulation for the entire world, we constructed the final product by selecting the best performing model for each region and variable. Global cross-validation correlations were ≥0.99 for temperature and humidity, 0.86 for precipitation and 0.76 for wind speed. The fact that most of our climate surface estimates were only marginally improved by use of satellite covariates highlights the importance having a dense, high-quality network of climate station data.

KEY WORDS interpolation; climate surfaces; WorldClim; MODIS; land surface temperature; cloud cover; solar radiation; wind speed; vapour pressure

For the **marine realm**, **Bio-ORACLE provides** uniform resolution, global coverage, and biologically significant variables, incorporating also CMIP6 Earth System Models and Shared Socioeconomic Pathways (SSPs).

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DATA ARTICLE

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Bio-ORACLE v3.0. Pushing marine data layers to the CMIP6 Earth System Models of climate change research

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Abstract

Motivation: Impacts of climate change on marine biodiversity are often projected with species distribution modelling using standardized data layers representing physi-


This helps evaluate climate models, leads to improvements in the model simulations and provides a better understanding of past, present and future climates.

State-of-the-art

Layers for future conditions comprise numerous Earth System Models.

ESM are **simulations that resolve global atmospheric and oceanographic conditions** across space and time (e.g., end-of-century SST in Culatra island).

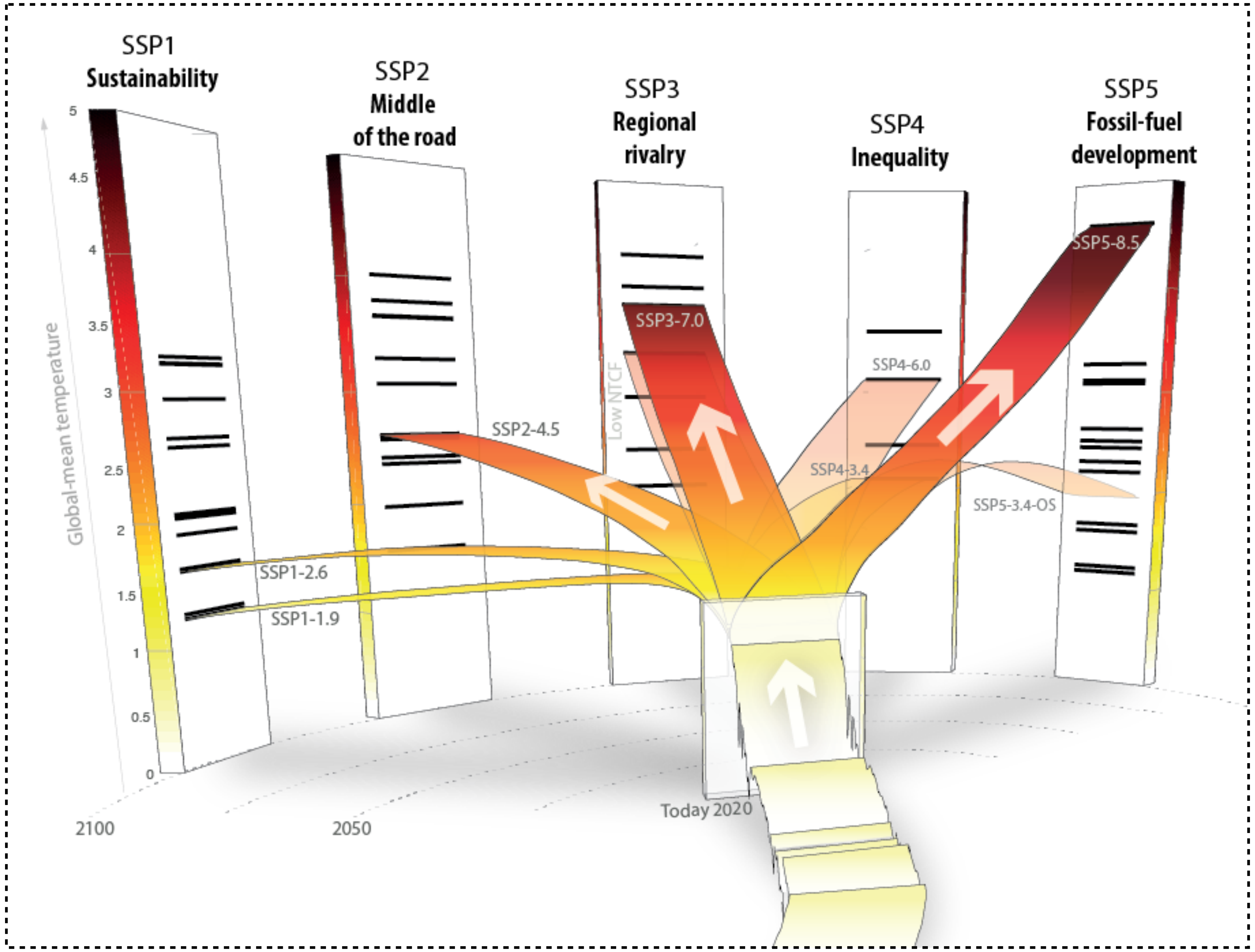
ESM **have inherent uncertainties**.

The **multi-model ensemble** (e.g., averaging different models) to **account for inter-ESM variability**. Previous available datasets limited to 2 ESMs (from the dozens available), limiting the potential to capture the range of climate change.

State-of-the-art

Layers for future conditions **consider data (ESM) from the 6th phase of the Coupled Model Inter-comparison Project.**

CMIP6 (1) is **more accurate in capturing the spatial and temporal variability of oceanographic conditions** and (2) **outputs the SSP scenarios**, which are the new narratives of global change.



The new **SSP use up-to-date socioeconomic assumptions** and **encompass the Paris Agreement expectations**, allowing the development of **projections aligned with international climate policies.**

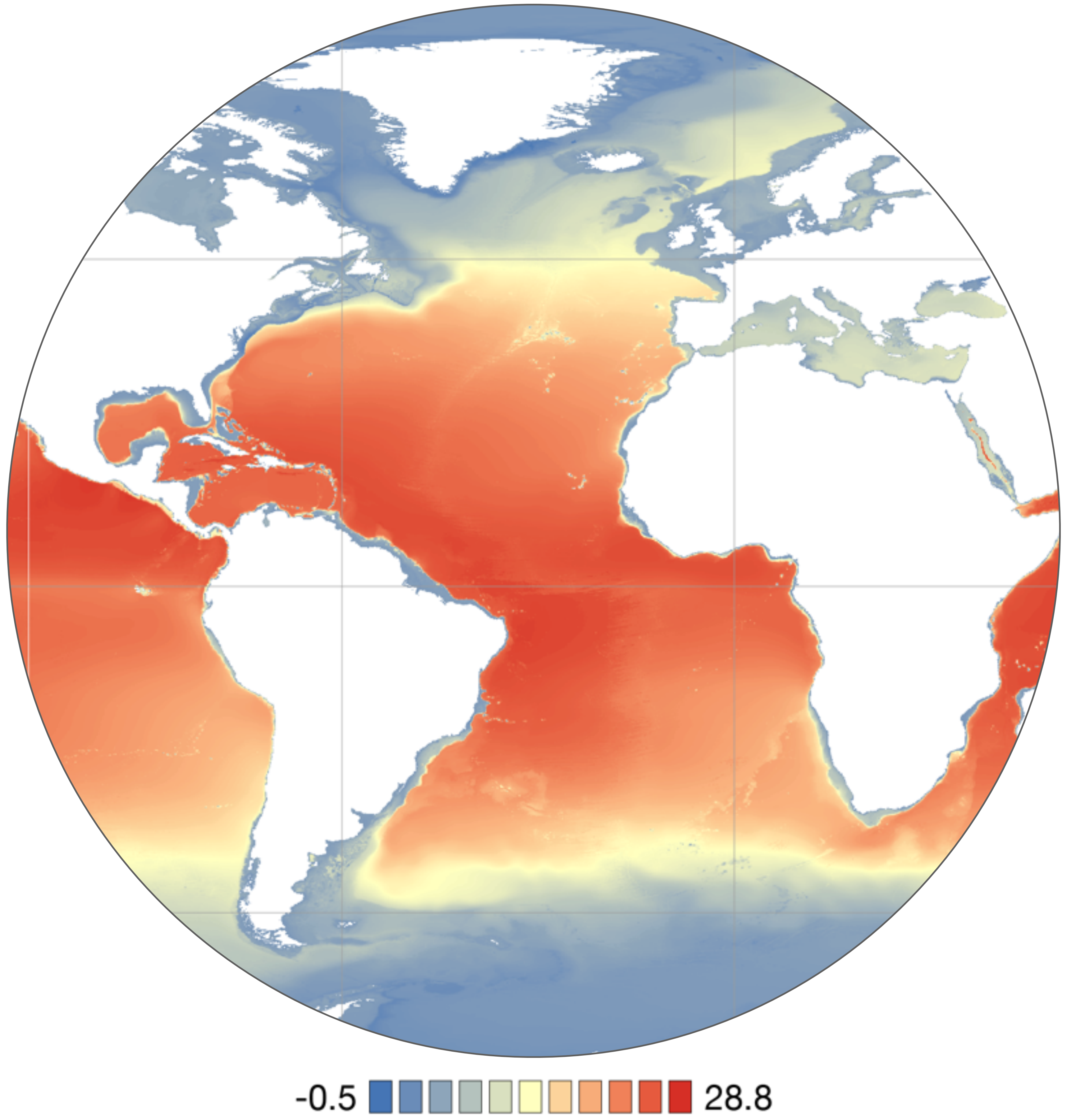
State-of-the-art

Moving beyond just temperature: While temperature is vital, relying on it exclusively misses the full physiological niche. Biologically meaningful models must incorporate variables like dissolved oxygen, primary productivity, and pH.

Layers for future conditions **consider biologically meaningful variables beyond temperature, salinity and sea ice conditions**.

Oxygen, primary productivity and pH (among others) **are limiting factors for biodiversity**, and are expected to change in the future.

Failing to consider these variables in climate change projections might **misinterpret the exposure of biodiversity to future detrimental conditions**.



Difference between surface and bottom temperature

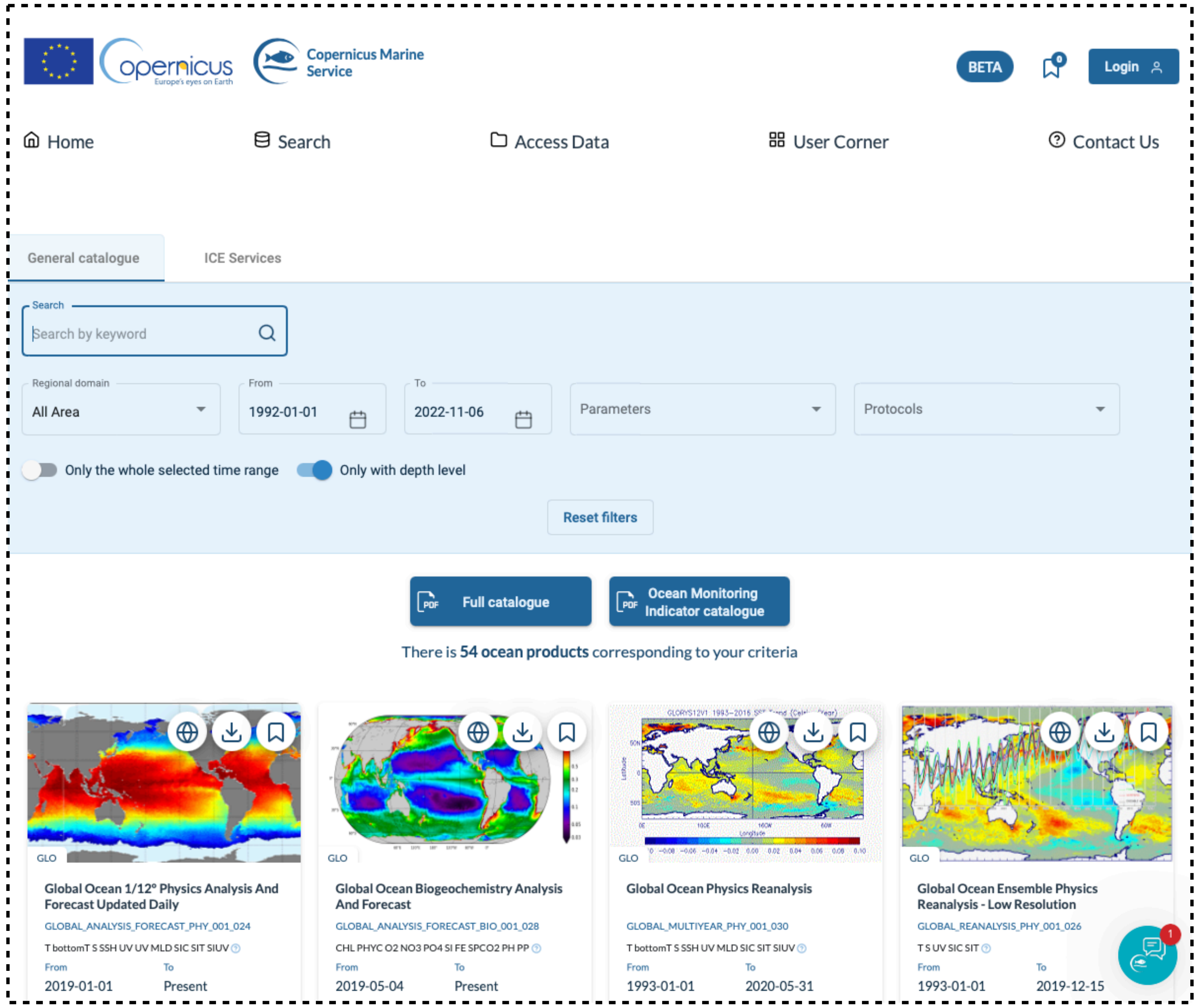
The 3D Ocean: Unlike terrestrial models, marine species are exposed to depth gradients. Sea bottom layers are key for **benthic species** (e.g., corals or fish).

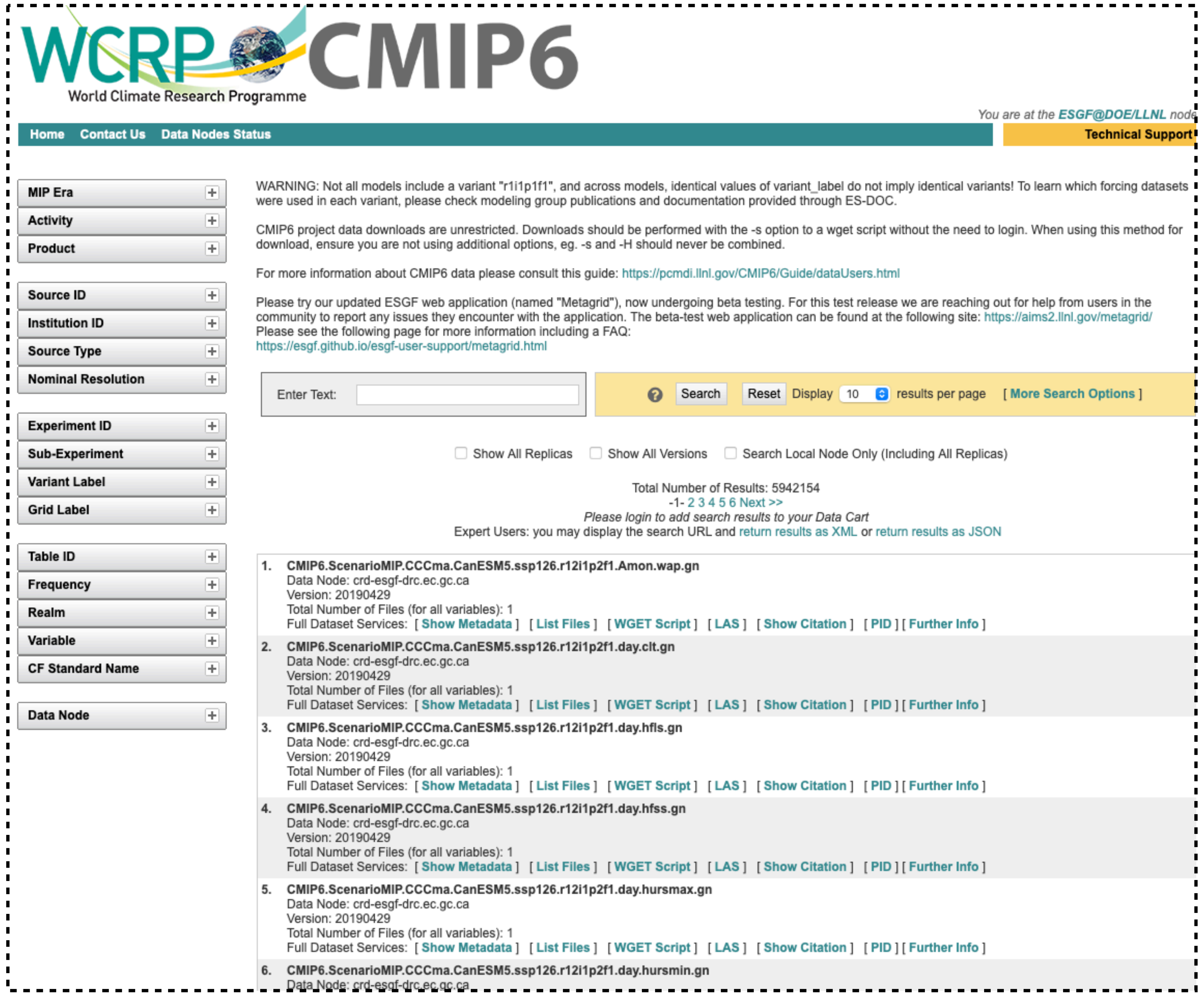
		Present	SSP1-1.9	SSP1-2.9	SSP2-4.5	SSP3-7.0	SSP4-6.0	SSP5-8.5
		1 (1970-2020)	4 (2020-2100)	4 (2020-2100)	4 (2020-2100)	4 (2020-2100)	4 (2020-2100)	4 (2020-2100)
1	**Air temperature**							
2	**Precipitation**							

WorldClim 2.1 (CIMP6): 2 terrestrial bioclimatic variables from 1970-2100, by 20-year averages at **0.5 arc-minutes resolution** (~1km), **covering the span of SSPs**, by **ensembling 23 GCMs of CMIP6.**

		Present	SSP1-1.9	SSP1-2.9	SSP2-4.5	SSP3-7.0	SSP4-6.0	SSP5-8.5
		2 (2000-2020)	8 (2020-2100)	8 (2020-2100)	8 (2020-2100)	8 (2020-2100)	8 (2020-2100)	8 (2020-2100)
1	**Temperature**							
2	**Salinity**							
3	**Sea Ice Cover**							
4	**Sea Ice Thickness**							
5	**Sea Water Velocity**							
6	**Mixed Layer Depth**							
7	**Diffuse Attenuation Coefficient**							
8	**PAR**							
9	**PAR at Bottom**							
10	**Oxygen**							
11	**pH**							
12	**Iron**							
13	**Phosphate**							
14	**Nitrate**							
15	**Silicate**							
16	**Primary Production**							
17	**Phytoplankton**							
18	**Chlorophyll**							

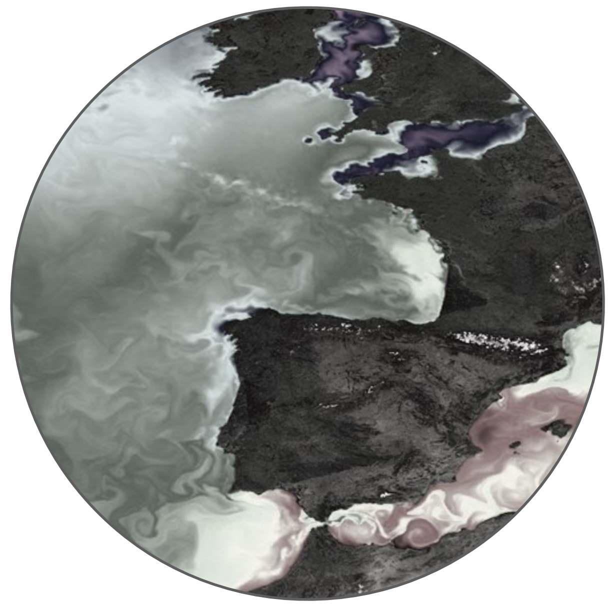
Bio-ORACLE provides **18 essential variables** provided for the surface and along the seafloor, **from 2000 to 2100, per decade**, at **0.05º resolution** (~5km), **covering the span of SSPs**, by **ensembling 10 ESMs of CMIP6**.





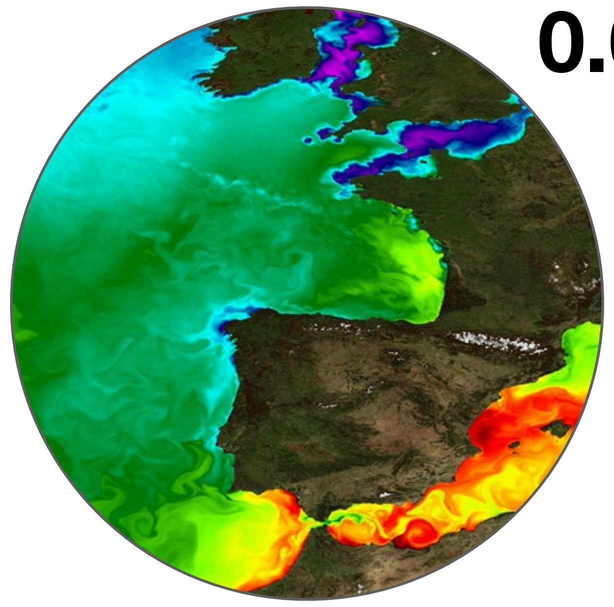
Examples of Data Sources:

Copernicus Marine Service and World Climate Research Programme



Raw data
1993-2025

Subsetting data and
resampling to desired resolution



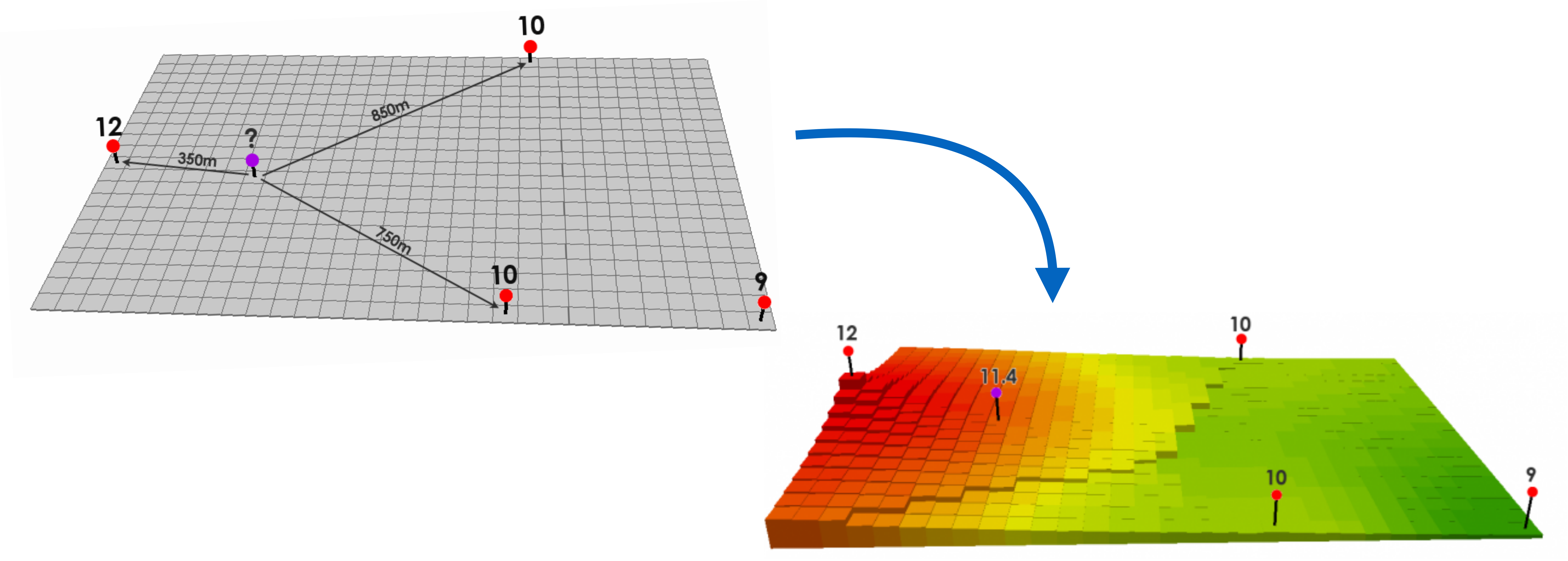
0.05º

Predictor Layer
2010-2020

For present-day conditions, layers are developed by subsetting raw data to the **desired decade** and **resampling** (interpolating) to the desired resolution (0.05º).

e.g.,

layer 1: Global maximum sea surface temperature of 2010-2020



Interpolation

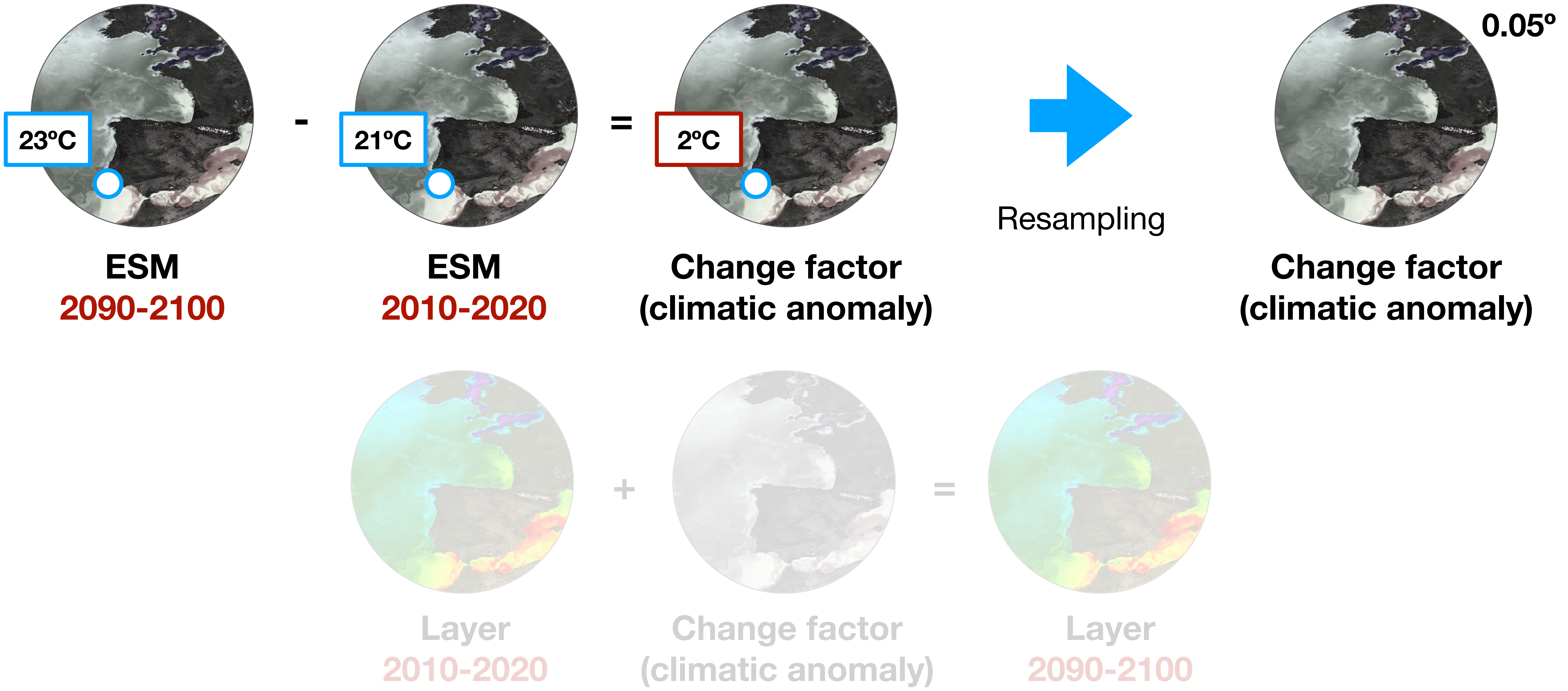
Raw **data are collected** from numerous sources (e.g., buoys, weather stations) and their information are spatially interpolated to infer the distribution of the environmental gradients for the whole region - **estimates unknown points**.

e.g., Inverse Distance Weighting algorithm.

For the future, use of the **change-factor technique**, applying the magnitude of climate change (projected by ESMs) to the present-day layers.

e.g.,

layer 2: Global maximum sea surface temperature of 2090-2100



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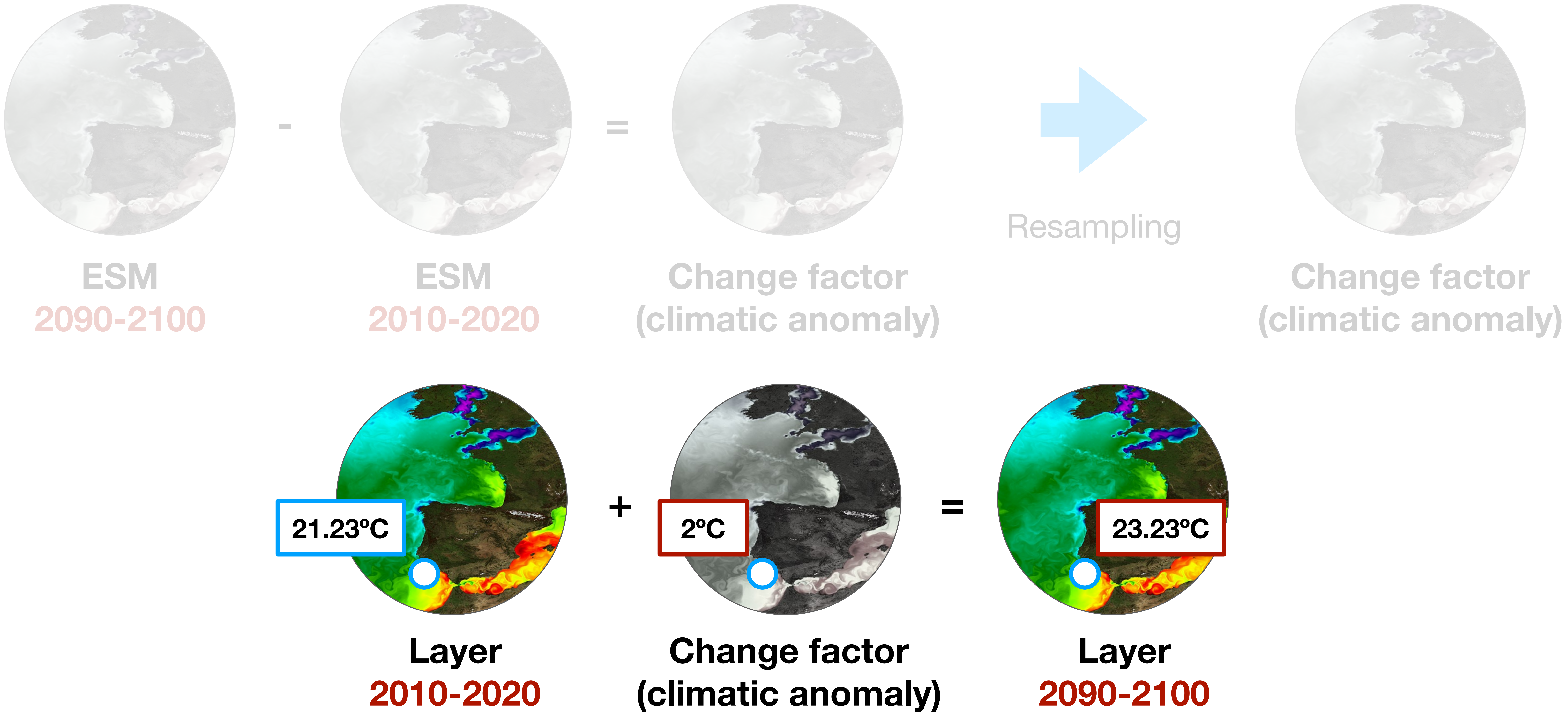

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layer 2: Global maximum sea surface temperature of 2090-2100

Literature

19. Bio-ORACLE v2.0 Extending marine data layers for bioclimatic modelling

20. Bio-ORACLE v3.0. Pushing marine data layers to the CMIP6 Earth System Models of climate change research

53. Kelp forest diversity under projected end-of-century climate change

57. BIOCLIM: the first species distribution modelling package, its early applications and relevance to most current MAXENT studies.

58. The Scenario Model Intercomparison Project (ScenarioMIP) for CMIP6

59. A fine-tuned global distribution dataset of marine forests